

Embedded system (Note)

- Embedded system is a dedicated computer system to perform a specific task. and control the interactions with its physical environment.
- embedded processes are typically integrated within or a larger mechanical or electrical systems
- Components of embedded system

① Micro controller / micro processor

- the brain of the system
- it execute instruction and process the data.

② Memory

- store the embedded software and data.

③ input device.

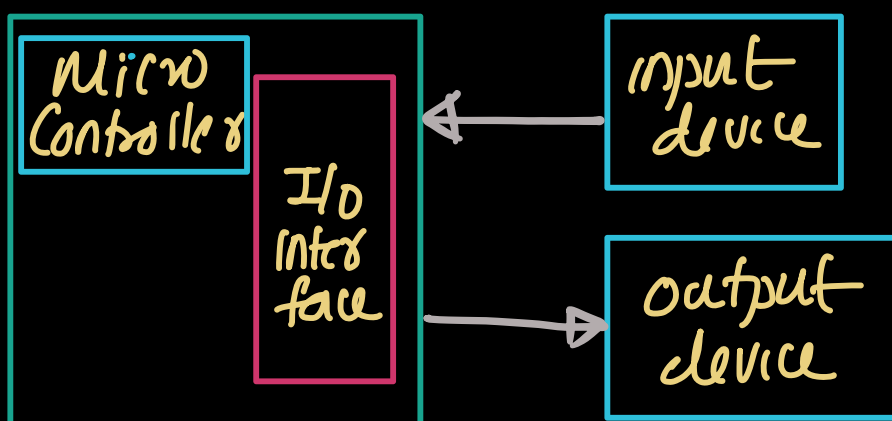
- components that gather data from the external environment and send it to system.
- eg: temperature sensor

④ Output device

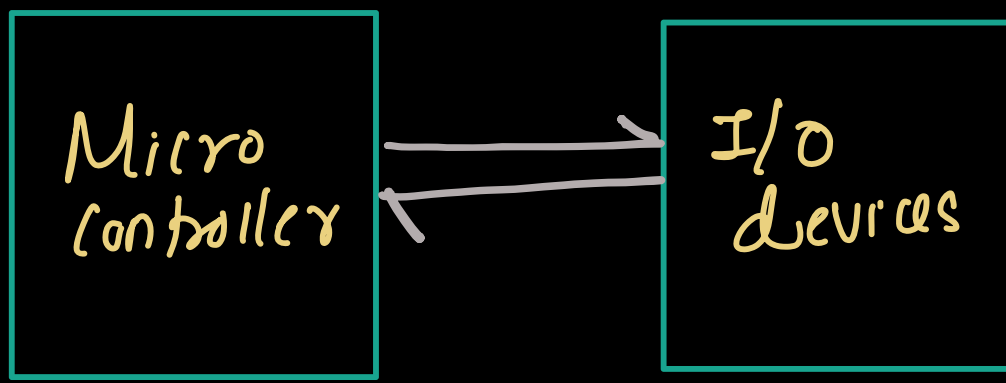
- components that receive processed data from the system
- eg: LCD display, buzzers

⑤ I/O interface

- Bridge circuits that allow the processor to communicate with external input output devices



Embedded I/O systems



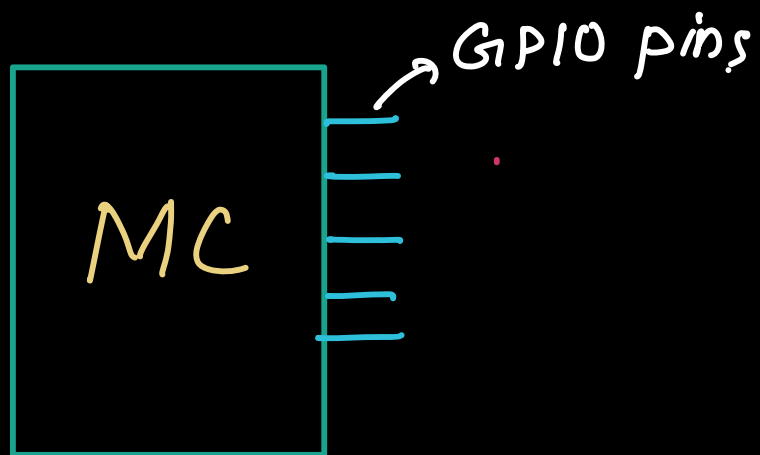
→ embedded I/O system is an input-output system used in the embedded system to communicate between microcontroller and external devices.

→ it enables embedded system to receive input from input devices and send output to the output devices.

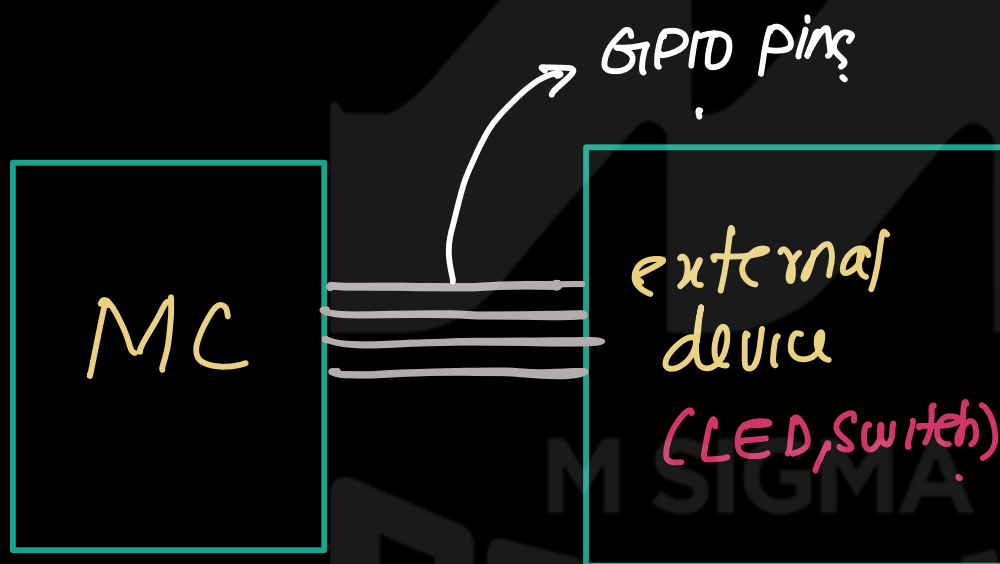
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General purpose I/O pins (GPIO pins)

→ General purpose I/O (GPIO) pins are physical pins present on the microcontroller that are used to read/write digital signals (logic 0 & logic 1)



→ These pins are used to enabling communication b/w the microcontroller and external devices such as LED, switch etc.



→ Each GPIO pins can be configured to operate either input mode or output mode depending on the requirement of the system.

— Input mode

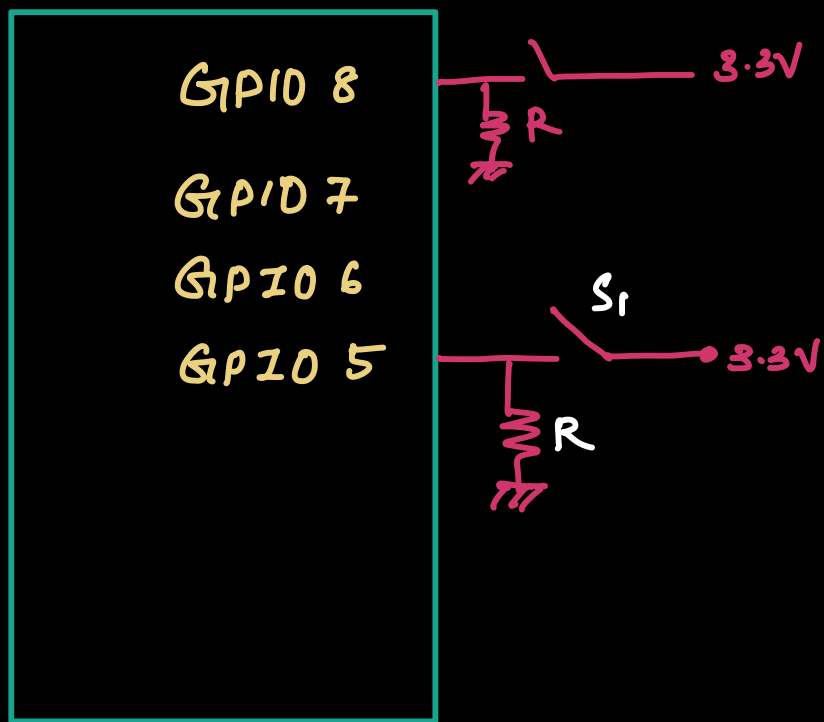
— In input mode, the GPIO pin is used to read signals from the external devices.

— In input mode GPIO pins reads the external voltage levels and interprets them as binary values (0 or 1)

If applied voltage is close to 0V, it is considered as Logic '0'

If applied voltage is sufficiently high (3.3V/5V) it is considered as logic '1'

- Example for input operation.
a switch is connected to GPIO pin (GPIO 5)



- When switch is open no voltage is applied to the pin hence signal at GPIO pin is Logic 1
- When switch is closed, voltage is applied to the pin hence signal at GPIO pin is Logic 1
- Pull down Resistor (R) is used to pull down the voltage at the pin when switch is open.

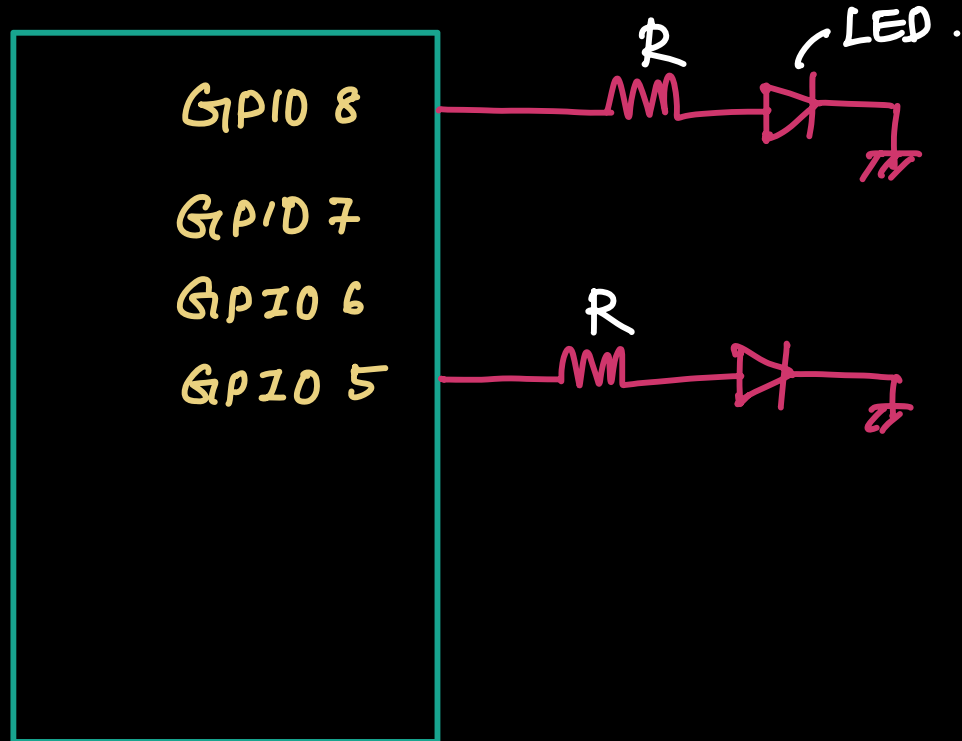
Output Mode.

- In output mode GPIO pins are used to send signal from microcontroller to external device.
- In this mode, microcontroller actively makes the pin to either Logic '0' or Logic 1
- When output is set to Logic 1, pin provides a voltage of 3.3V/5V and when it is set to Logic 0, the pin is connected to ground.

→ Example : LED is connected to GPIO pin.

— When GPIO pin is at Logic high, then potential difference is created and current flows through the LED, LED turns ON.

→ When GPIO pin is at Logic '0', No current flows through the LED, LED remains in off state.



If the signal at the GPIO pin (GPIO 5) is logic 1

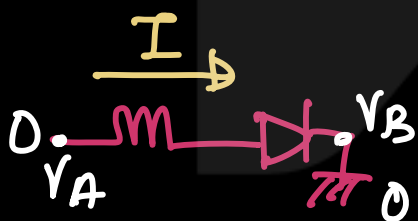


$$V_A = 5V$$

$$V_B = 0V$$

$\rightarrow$ So there is a potential difference
so current flows through the diode

If the signal at the GPIO pin (GPIO 5) is logic 0



$$V_A = 0$$

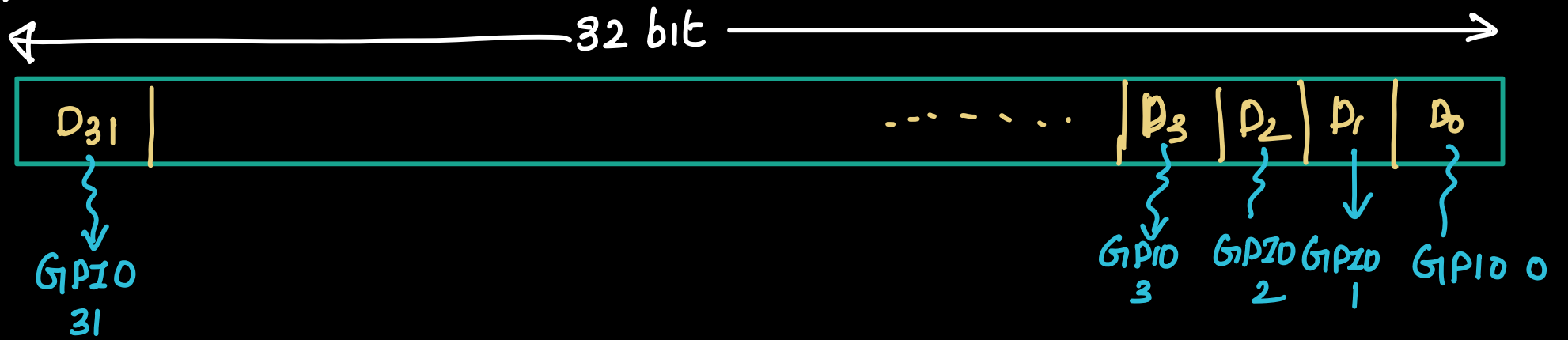
$$V_B = 0$$

No voltage difference, so $I = 0$.

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GPIO Registers

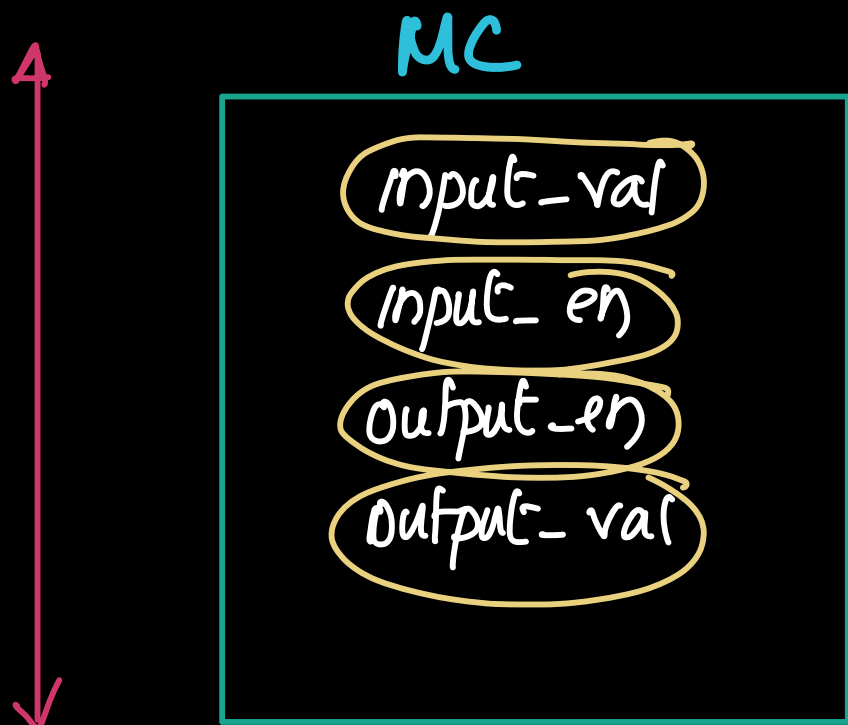
- the operation of GPIO pins is controlled through special Registers (Memory mapped I/O Register)
- Each Register are 32 bit wide, each bit corresponds to one bit in these registers.



Registers

- To perform input and output operation, different registers are used

- ① input_val (input value) Register
To read the data from the pin
- ② input_en (input enable) Register
To configure pins as input
- ③ output_en (output enable) Register
To configure pins as output
- ④ Output_val (output value) Register
To write data to the pins



these Register control the operation of GPIO pin

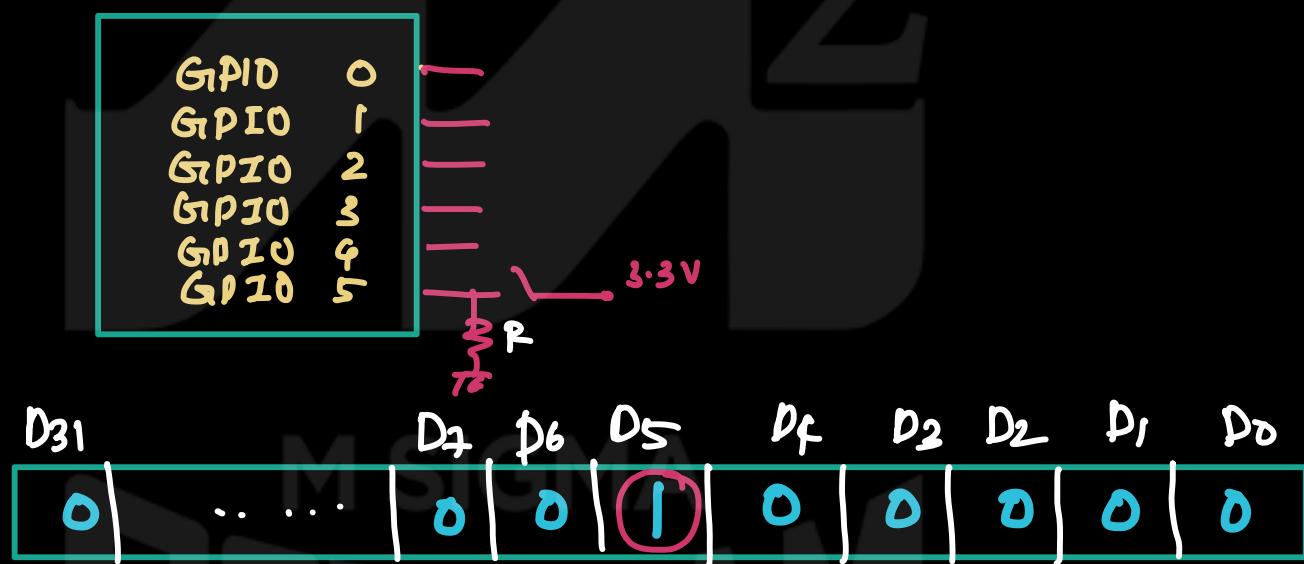
Operation of GPIO - Reading from GPIO pin

- Read the data from GPIO pin to the microcontroller
- The process of reading data from a GPIO pin involves the following steps.

① Configure the GPIO pin as an input pin

Before read the data, GPIO pin must be configured as an input pin, this is done by setting corresponding bits in the input-en register.

Eg: If a switch is connected to GPIO pins. then make sth bit of input-en register to 1

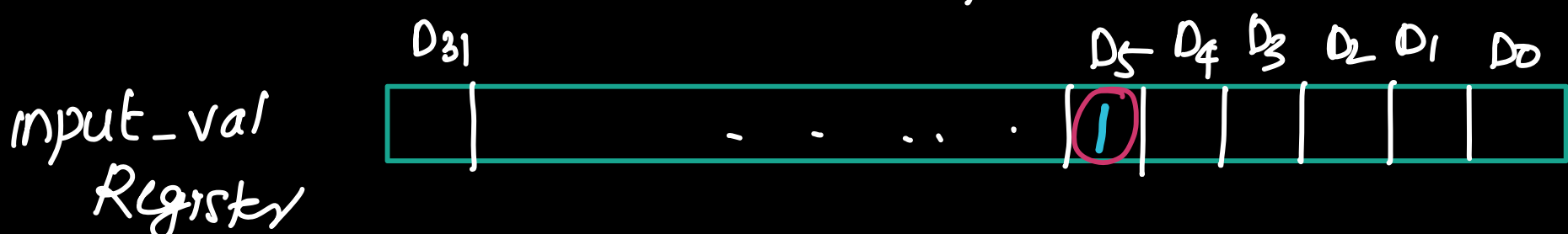


② reading the input value

GPIO hardware continuously monitors the voltage level at the pin. and store the corresponding digital value in the input value register.

eg: when switch closed, voltage at the pins is high, corresponding bit in the input-val Register become one.

when switch is opened, voltage at the pin is '0', so corresponding bit in the input-val register become zero



③ extracting required bit.

To determine the state of a specific pin, processor checks the corresponding bits in input value register.

eg: If 5th bit = 1, input is high.
If 5th bit = 0, input is low.

Operation of GPIO pin - Output Mode

- In embedded system, GPIO pins are used to send digital signals to the external devices such as LEDs...
- In this case pin must be configured as output pin and then write the value in output value register.

Steps:

steps:

- ① Configuring GPIO pin as output pin

Before sending any signal, GPIO pin must be configured as an output pin. This is done by setting the corresponding bit in the output enable register.

eg: Consider an LED is connected to GP205
then set 5th bit of output-en Register
to 1.

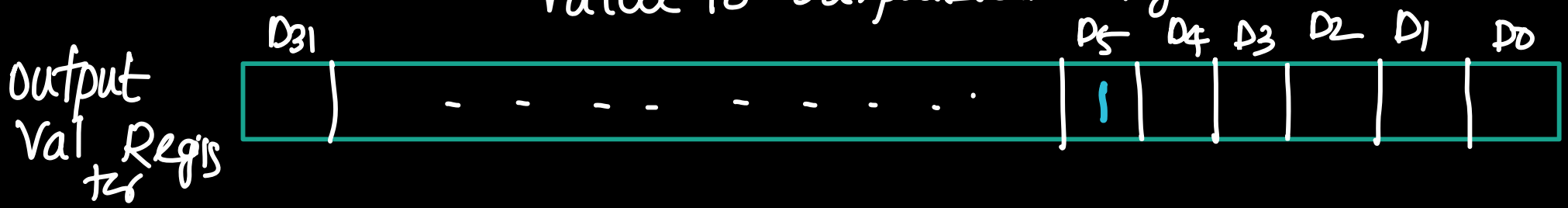
② Writing output value

after enabling the pin as output, the required signal is send by writing to the output value register.

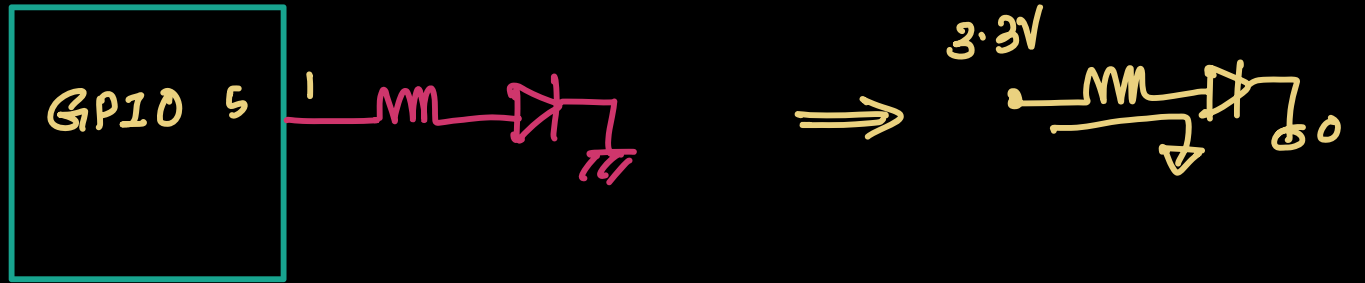
If bit $n=1$, then n^{th} pin output is high (3.3V)

if bit $n = 0$, the n^{th} pin output is low (00)

eg: If LED is connected to GPIO 5, then to send a value to GPIO 5, write the value to output_val Register.



- ③ Apply signal to the external devices.
- The voltage generated at the GPIO pin is applied to an external device.
 - When output is high, a potential difference is created, allowing current through the device (LED). external device responds based on the signal (LED turns ON)



- When output is low, there is no potential difference, so no current flows through the device and the external device responds based on the signal (LED turns OFF)

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